

Followup 26-09

Unresponsive with an unequal pupil after a reported seizure

Patient	77-year-old female, resident of a skilled nursing facility for the preceding two weeks. Cerebral meningioma treated with radiation, with ongoing vasogenic edema of the frontal lobes, left worse than right. Seizure disorder dating to three months earlier, coronary artery disease, hypertension, dysphagia. On levetiracetam, and started on lacosamide two days before this call.
Outcome	Survived. CT found no bleed, no infarct, and a meningioma unchanged from previous imaging. An EEG the next day found no seizure activity at all, only the slowing and triphasic waves of a diffuse encephalopathy. Both of her anticonvulsant levels came back supratherapeutic , and she had a urinary tract infection growing two organisms. The discharge summary resolved supratherapeutic anticonvulsant level and urinary tract infection, and listed the admission diagnosis as altered mental status.
Disposition	Eight days inpatient. Both anticonvulsants held and restarted at lower doses, ceftriaxone for the infection, a catheter for urinary retention with a urology referral. Discharged back to the facility still minimally interactive, with outpatient neurology and an EEG arranged.

The call

Dispatched priority 2 for a 77-year-old with a seizure history who had seized and was no longer seizing.

They found her seated in her wheelchair in her room. **Her eyes were open and she was not alert**, with no response to verbal or painful stimulus. Pale, warm, and dry, with a strong regular radial pulse around 70 and respirations around 14. She was drooling, which staff said was not normal for her.

The history the crew got on scene is the most valuable thing in this record. Staff said they had been in her room about fifteen minutes earlier when she suddenly **"got real stiff but did not convulse,"** and that she had been unresponsive since. No falls, no trauma. **The nurse believed her seizure medication had recently been adjusted.** And staff volunteered that all of her previous seizures had looked very different, and that they were not certain what they had seen was a seizure at all.

Course, relative to patient contact:

- **On contact:** Airway, breathing and circulation assessment, neurologic assessment, Cincinnati Prehospital Stroke Scale, stroke screen. Vascular access attempted.
- She was lifted from the wheelchair to the stretcher. **During the move, her respirations became slower, deeper, and labored, and pulse oximetry could not be obtained.**
- **+5 min:** Non-rebreather at 15 L. Respirations returned to normal over two to three minutes.
- **+7 min:** 160/92, MAP 115, pulse 76, respirations 16, SpO2 99%. Cardiac monitor applied.
- Non-rebreather removed once she was breathing evenly and unlabored. Capnography cannula placed, saturation 94 to 99% on room air, no cyanosis.

- **+14 min:** 133/88, MAP 103, pulse 78, respirations 25, SpO2 94%. **Glucose 235.**
- En route, vascular access attempted twice more without success. **Her left pupil was noted to be larger than the right and non-reactive to light. The right was round and reactive.** She was incontinent of urine.
- The receiving facility was notified by phone of an incoming possible stroke, and asked for a room on arrival.
- **+22 min:** Bedside handoff.

She remained unresponsive throughout, unable to follow commands, and the stroke scale could not be completed. No obvious facial droop.

Two things to keep in mind throughout the review: **The pupil**, and **the sentence about her medication being adjusted.**

Primary impression

Unresponsive, possible stroke, with an unequal pupil.

That is the right impression and it drove the right actions. A new anisocoria in an unresponsive patient is a structural emergency until a scanner says otherwise, and the crew called ahead as a possible stroke and asked for a room.

The scanner said otherwise. No hemorrhage, no infarct, ventricles and sulci and cisterns all normal, and the meningioma unchanged from prior imaging. By the time neurology examined her the following morning, **her pupils were equal, round, and reactive bilaterally.** The anisocoria was real, it was documented, and it resolved without anyone doing anything about it.

What actually made her unresponsive took eight days to establish. Her levetiracetam came back at **55.0 against an upper limit of 46** and rose to **61.1** on repeat, with lacosamide also elevated. She had been on a full dose of one drug while a second was started two days earlier, with no documentation anywhere of what the outpatient plan had been, and a urinary tract infection growing two organisms on top of it.

She probably did not have a seizure. The facility staff who were standing in the room said as much to the crew, and the crew wrote it down.

Hospital course

Emergency department, day 0. She arrived opening her eyes intermittently but not following commands, protecting her own airway, with wet clothing from the incontinence. Triage 137/84 with a pulse of 80. Glucose 232. The working diagnosis was a breakthrough seizure with a postictal state, and she was started on a levetiracetam load and a broad workup: CT head, blood counts, chemistry, urinalysis, drug screen, levetiracetam and lacosamide levels, acetaminophen, salicylate, and ethanol.

CT head found no acute hemorrhage and no infarct, with a 3.9 cm partially calcified meningioma in the planum sphenoidale region and anterior falx and associated vasogenic edema, left more than right, unchanged from imaging three weeks earlier. The ventricles, sulci, and cisterns were normal. Documented as not consistent with stroke and not a thrombolysis candidate.

Urinalysis was turbid with positive nitrite, large leukocyte esterase, more than 100 white cells per field, and many bacteria, and ceftriaxone was given. The levetiracetam level resulted at **55.0** and the loading dose that had been started was stopped. Chart review found she had been admitted three weeks earlier for a seizure, discharged on levetiracetam and a dexamethasone taper for the edema around her treated lesion, and had completed the taper.

Day 1. EEG. No seizures and no epileptiform discharges. What it did show was moderate generalized slowing, generalized rhythmic delta activity, left temporal lateralized rhythmic delta activity, and **generalized periodic discharges with triphasic morphology**, read as a diffuse encephalopathy from a metabolic, toxic, infectious, or other systemic cause.

Days 2 through 8. She remained obtunded and poorly interactive. Urine culture grew *Escherichia coli* and *Proteus mirabilis*. The repeat levetiracetam level came back at **61.1**, higher than the first. Neurology held both anticonvulsants for two doses and restarted them at reduced doses. Improvement was slow and incomplete. She began eating again and was awake and feeding herself by the last full day, and was discharged back to the facility on day 8 still with significant psychomotor slowing, with instructions for repeat levels in a week.

Final diagnoses: altered mental status; supratherapeutic anticonvulsant levels, resolved; urinary tract infection, resolved; history of seizures with prolonged postictal state; sphenoidal meningioma with vasogenic edema of the frontal lobes; urinary retention; thrombocytopenia.

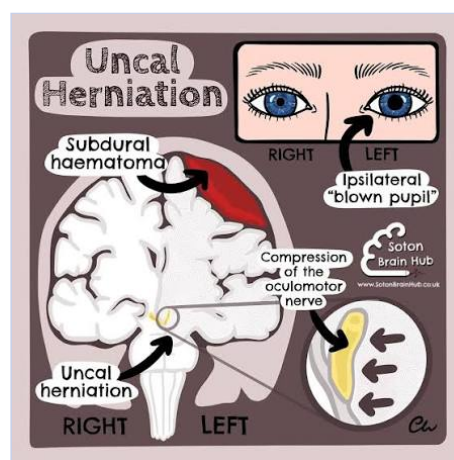
Education points

The unequal pupil

The crew found a left pupil that was larger than the right and did not react to light, in a patient who was unresponsive. That combination is one of the few findings in prehospital medicine that justifies moving immediately, and it is worth understanding exactly what it means.

Pupil size is a balance between two nerve supplies. Sympathetic fibers dilate it. **Parasympathetic fibers constrict it, and they travel on the outside surface of the third cranial nerve**, which runs from the midbrain forward to the eye. Sitting on the outside is the anatomical fact that makes all of this matter, because it means they are the first thing crushed when anything presses on that nerve. Lose them and the sympathetic supply is unopposed, so the pupil dilates and stops responding to light.

The nerve runs directly over the edge of the tentorium, the shelf of dura separating cerebrum from cerebellum. When pressure in one half of the skull rises far enough, the innermost part of the temporal lobe is pushed over that edge onto the nerve. That is **uncal herniation**, and a fixed dilated pupil on the same side as the pressure is its earliest reliable sign. A new unequal pupil in an unresponsive patient is the brain telling you it has run out of room.



Uncal herniation and the ipsilateral blown pupil. The diagram shows the general mechanism, with a subdural hematoma as the example cause. This patient had no herniation and no subdural.
Illustration: Soton Brain Hub.

Hers was not that. Her CT showed normal ventricles, sulci, and cisterns, no midline shift, and no change in her meningioma. Her pupils were equal and reactive by the next morning. Nobody ever explained the finding, and the record does not try to. What the record does contain is a left-sided pattern that lines up: her vasogenic edema is left worse than right, and her EEG showed left temporal and left hemispheric slowing greater than the right.

There are benign explanations. Physiologic anisocoria is present in roughly one person in five, an anticholinergic reaching one eye will dilate it, and a postictal or focally dysfunctional hemisphere can produce transient asymmetry without any mass effect. **None of that changes what you do, because you cannot tell them apart at the bedside and one of them kills your patient inside of an hour.** Treat a new unequal pupil as herniation, move, and let the donut of truth sort it out. That is what this crew did.

Further reading on anisocoria. The full differential, including the pharmacologic and benign causes, goes further than a case document can carry. StatPearls has a free chapter: <https://www.ncbi.nlm.nih.gov/books/NBK470384/>

Too much medicine looks like pathology

Her levetiracetam level was 55.0 against a range topping out at 46, and it went up to 61.1 on a repeat draw. Her lacosamide level was also high. Both are anticonvulsants, and both work by damping down neuronal excitability. Push that far enough and you do not get a calmer brain, you get a slower one.

Levetiracetam reduces the release of excitatory neurotransmitter. Lacosamide slows the recovery of sodium channels so neurons cannot fire repetitively. **Different mechanisms aimed at the same target**, which is why they are used together and why stacking them sedates additively. At supratherapeutic levels the expected findings are exactly what she had: somnolence, slowed responses, and poor interaction.

The route by which she got there is the part worth clarifying. She had been on levetiracetam. A second drug was started two days before this call. The chart contains no documentation of what the outpatient neurologist actually intended, and the notes record the facility reporting that the family had been told they were weaning the levetiracetam while a nurse separately reported an unknown dose. **A cross-titration between two drugs is the single most dangerous week in a seizure patient's medication history**, because for that week the patient is on both, and any failure of communication means they are on full doses of both.

Her kidney function was normal, so she was not failing to clear the drug. She was simply taking more of it than anyone intended.

The EEG confirmed it. Across a full hour it found no seizures and no epileptiform activity, only generalized slowing and **triphasic waves**, which are the classic electrical signature of a toxic or metabolic encephalopathy. That is the pattern of a brain being poisoned rather than one misfiring, and in a patient admitted for a seizure it points the diagnosis in the opposite direction.

Prehospital considerations

The history from the people in the room is the diagnosis here. Three things the staff said turned out to matter more than anything the crew could measure. That she **got stiff but did not convulse**. That **all of her previous seizures looked different and they were not sure this one was a seizure**. And that **her seizure medication had recently been adjusted**. The last of those is the answer to the whole case, and it came from a nurse saying she believed it rather than knowing it. The crew wrote it down as belief rather than fact, which is the correct way to carry information you cannot verify.

Dispatch had already labelled this a seizure and the crew did not accept the label. The complaint came in as a seizure in a patient with a seizure history. Everything downstream, including the emergency department's

working diagnosis, ran on that label; but it is worthwhile to note that they were only able to discount the working theory of an intercranial hemorrhage because they had the labs and scans to prove it. The crew did a good job avoiding anchor bias, maintaining a wide differential, and acting on the most severe working diagnosis that fit the presentation.

A breathing pattern that changes during a move is a finding, not a handling problem. Her respirations became slower, deeper, and labored during the transfer, and pulse oximetry could not be obtained. Slow and deep in an unresponsive patient is not the fast shallow pattern of respiratory distress; it is a brainstem pattern, and in a patient with a possible mass it is the third element of the Cushing response, after the raised pressure and the relatively slow heart rate she also had at 160 over 92 with a pulse of 76. The crew put oxygen on, watched it resolve over two to three minutes, then reassessed and weaned it. That is the right sequence in both directions.

Capnography is the right instrument for a changing respiratory pattern. It is the only field measurement that shows ventilation rather than oxygenation, and it would have told the crew whether that slow deep breathing was moving air. A single value of 16 imported into the EMS chart. Remember that the numbers porting into the events tab are only quantitative measurements, for elements like Capnography waveforms, ecg tracings, and pulse oximetry pleths a description of the waveform in the narrative can provide more insight, especially if you're making decisions on that information.

Check the glucose. Two readings, 235 in the field and 232 in the department. Hypoglycemia is the one cause of an unresponsive patient you can find and fix in sixty seconds.

The interventions

PREHOSPITAL INTERVENTIONS

Stroke scale attempted and documented as incomplete. The Cincinnati scale requires the patient to smile, hold their arms out, and speak, and she could do none of it. Documenting that a scale **could not be completed** is a finding rather than a gap, because a score of zero would hide the depth of impairment. An unresponsive patient is not a negative stroke screen.

Oxygen by non-rebreather at 15 L, then removed. Given for a changing respiratory pattern and an unobtainable saturation, which is the right threshold. The instructive half is the removal, once she was breathing evenly and saturating 94 to 99% on room air. **Oxygen is a drug with a dose**, and taking it back off is as much a decision as putting it on.

Vascular access attempted and not obtained, on scene and twice more en route. The hospital likes to have an 18ga in the right arm, at or proximal to the antecubitum. The crew did a good job balancing that request with the desire not to delay transport for a line they were not going to use themselves.

Notification by phone as a possible stroke, with a request for a room. The most valuable intervention on this call. An unresponsive patient with a new unequal pupil needs a scanner and a physician next to it, and that only happens if somebody calls while still moving.

IN-HOSPITAL INTERVENTIONS

CT head without contrast. The first question in an unresponsive patient with a focal finding is whether there is blood or a mass pressing on something, because those have time-critical answers. Non-contrast CT is fast and it is very good at exactly that question. Hers was negative for both, and negative in a specific and useful way: **normal ventricles, sulci, and cisterns**, which is the radiologist saying there is no pressure problem.

Continuous EEG. An hour of recording with video, looking specifically for **nonconvulsive status epilepticus**, which is a brain still seizing electrically in a body that has stopped moving. In a patient who does not wake up when she should have, that has to be excluded before anything else is accepted. Hers excluded it and found the toxic-metabolic pattern instead.

Holding both anticonvulsants, then restarting lower. The treatment for drug toxicity is less drug, and in a patient with a genuine seizure disorder that is a calculated risk rather than an obvious move. Neurology held two doses of each and restarted both lower, accepting a short window of reduced coverage to get the levels down.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Anticonvulsant levels

Lab	Day 0	Day 4	Reference range
Levetiracetam, serum (mcg/mL)	55.0	61.1	12.0 to 46.0
Lacosamide, serum	Elevated, value not reported	not repeated	Therapeutic range per assay

These two numbers are the case. A level above the therapeutic range means the drug is accumulating faster than the body is clearing it, and the second value being higher than the first means it was still accumulating on day 4, after she had already been in hospital for several days.

A therapeutic range is the band in which most patients get the benefit without the side effects, and both edges matter. Below it the seizures are not controlled. Above it the sedating effects that are mild at normal doses become the whole clinical picture, which for her meant eight days awake and unable to interact.

The reason it matters to a crew is not that you will ever draw one. **A medication level is a question the hospital can answer and you cannot, and it only gets asked if somebody flags the medication change.** Nothing else in this presentation pointed at it.

The infection, and a normal white count

Lab	Value	Reference range
White blood cell (x10 ⁹ /L)	9.3	4.0 to 10.8
Neutrophils (%)	82.4	35.0 to 80.0
Lymphocytes (%)	6.4	20.0 to 50.0
Lymphocytes, absolute (x10 ⁹ /L)	0.59	1.00 to 4.00
Immature granulocytes, absolute (x10 ⁹ /L)	0.08	0.00 to 0.05
Urine nitrite	Positive	Negative
Urine leukocyte esterase	Large	Negative
Urine white cells (per high power field)	More than 100	0 to 5
Urine bacteria	Many	None

Her total white count was entirely normal at 9.3, and she had a florid urinary tract infection. The urine had more than a hundred white cells per field, many bacteria, positive nitrite, and grew two separate organisms. If you were waiting for a raised white count to believe in the infection, you would have missed it.

The finding is in the differential rather than the total. Her neutrophils were 82.4% against a ceiling of 80, her lymphocytes were 6.4% against a floor of 20, and her absolute lymphocyte count was 0.59 against a floor of 1.0. **That combination, neutrophils up and lymphocytes down, is the stress and infection pattern,** and it is entirely hidden inside a normal-looking total because the two shifts cancel each other out. Her immature granulocytes were also just above range, which is the marrow releasing cells early.

Older adults frequently do not mount a white count or a fever in response to infection, and in this patient there is a second reason: she had recently completed a course of dexamethasone, and steroids blunt exactly this response. Infection in an elderly patient often presents as confusion with normal vital signs, and the laboratory does not necessarily rescue you either.

Everything else was checked and was normal. Glucose at 232, sodium, kidney function, thyroid, acid-base, alcohol, salicylate, acetaminophen, and a urine drug screen all came back clean, which is what left the medication levels and the infection standing.

Learning points

1. **A new unequal pupil in an unresponsive patient is herniation until a scanner says otherwise.** The parasympathetic fibers that constrict the pupil ride on the outside of the third cranial nerve, so they are crushed first when the temporal lobe is pushed over the tentorium. You cannot tell that apart from a benign cause at the bedside, and one of them will kill your patient quickly. This crew found it, called ahead as a stroke, and asked for a room, which is right even though the CT turned out clean.
2. **Ask what changed in the medications, and write down who told you.** A nurse believing the seizure medication had recently been adjusted is the sentence that explains this entire admission. She was on a full dose of one anticonvulsant with a second added two days earlier, and both levels came back supratherapeutic. Nobody could produce the actual plan. A cross-titration between two drugs is the most dangerous week in a seizure patient's year.
3. **When the people who watched it say it did not look like a seizure, that is data.** Staff told this crew she got stiff without convulsing, that her previous seizures looked nothing like it, and that they were not certain what they saw. Dispatch had already called it a seizure. A full hour of EEG found no seizure activity at all. The account from the person standing in the room outranks the label on the call.
4. **A normal white count does not exclude infection in an older patient.** Hers was 9.3 with a urine full of bacteria and two organisms on culture. The finding was in the differential, with neutrophils above range and lymphocytes well below it, which is a real shift hidden inside a normal total. Age and recent steroids both blunt the response.
5. **Slow and deep breathing in an unresponsive patient is a brainstem sign, not a comfort problem.** Hers changed during the move, resolved over a couple of minutes on oxygen, and sat alongside a mean arterial pressure of 115 with a pulse of 76. Capnography characterises it, because it measures ventilation rather than oxygenation, and it is the number and waveform worth having when the pattern is what is changing.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at <https://bcmca.github.io/followup/>